DBMS Lab Assignment 6 - Solution

Table EMP with SELECT queries, grouping, date functions and constraint handling. Each question shows the Oracle SQL followed by its output.

*Note: outputs were produced by running equivalent statements on a SQLite database with the same table and data, formatted like SQL*Plus. SYSDATE is taken as 09-Nov-2026 (the run date).*

Create table EMP

SQL

```
CREATE TABLE EMP (
  EMP_ID     NUMBER(6)    CONSTRAINT pk_emp PRIMARY KEY,
  FIRST_NAME VARCHAR2(20) CONSTRAINT nn_emp_fname NOT NULL,
  LAST_NAME  VARCHAR2(25) CONSTRAINT nn_emp_lname NOT NULL,
  HIRE_DATE  DATE         CONSTRAINT nn_emp_hdate NOT NULL,
  JOB_ID     VARCHAR2(10) CONSTRAINT nn_emp_job   NOT NULL,
  SAL        NUMBER(8,2)  CONSTRAINT nn_emp_sal   NOT NULL,
  DEPT_ID    NUMBER(3)    CONSTRAINT chk_emp_dept CHECK (DEPT_ID >= 10));
```

OUTPUT

```
Table created.

SQL> DESC EMP
Name         Null?     Type
-----------  --------  ------------
EMP_ID       NOT NULL  NUMBER(6)
FIRST_NAME   NOT NULL  VARCHAR2(20)
LAST_NAME    NOT NULL  VARCHAR2(25)
HIRE_DATE    NOT NULL  DATE
JOB_ID       NOT NULL  VARCHAR2(10)
SAL          NOT NULL  NUMBER(8,2)
DEPT_ID                NUMBER(3)
```

Note: Column mapping from the sheet: EMP_ID = PK, the next five columns are NOT NULL, and the 'greater than or equal to 10' rule applies to DEPT_ID.

Insert the data

SQL

```
INSERT INTO EMP VALUES (198,'Donald','Connell',TO_DATE('21-Jun-99','DD-Mon-RR'),'SH_CLERK',2600,50);
INSERT INTO EMP VALUES (199,'Douglas','Grant',TO_DATE('13-Jan-98','DD-Mon-RR'),'SH_CLERK',3000,50);
INSERT INTO EMP VALUES (200,'Jennifer','Whalen',TO_DATE('17-Sep-87','DD-Mon-RR'),'AD_ASST',4400,10);
INSERT INTO EMP VALUES (201,'Michael','Hartstein',TO_DATE('19-Jan-99','DD-Mon-RR'),'IT_PROG',6000,20);
INSERT INTO EMP VALUES (202,'Pat','Fay',TO_DATE('25-Oct-89','DD-Mon-RR'),'AC_MGR',6500,20);
INSERT INTO EMP VALUES (203,'Susan','Mavris',TO_DATE('26-Nov-76','DD-Mon-RR'),'AD_VP',7500,40);
INSERT INTO EMP VALUES (204,'Hermann','Baer',TO_DATE('23-Aug-95','DD-Mon-RR'),'AD_PRES',9500,90);
INSERT INTO EMP VALUES (205,'Shelley','Higgins',TO_DATE('24-Feb-98','DD-Mon-RR'),'AC_MGR',2300,60);
INSERT INTO EMP VALUES (206,'William','Gitz',TO_DATE('12-Mar-01','DD-Mon-RR'),'IT_PROG',5000,60);
INSERT INTO EMP VALUES (100,'Steven','King',TO_DATE('15-Jun-02','DD-Mon-RR'),'AD_ASST',8956,100);
INSERT INTO EMP VALUES (101,'Neena','Kochar',TO_DATE('10-Jul-03','DD-Mon-RR'),'SH_CLERK',3400,30);
COMMIT;
```

OUTPUT

```
11 rows created.

EMP_ID  FIRST_NAME  LAST_NAME  HIRE_DATE  JOB_ID     SAL  DEPT_ID
------  ----------  ---------  ---------  --------  ----  -------
   206  William     Gitz       12-MAR-01  IT_PROG   5000       60
   205  Shelley     Higgins    24-FEB-98  AC_MGR    2300       60
   204  Hermann     Baer       23-AUG-95  AD_PRES   9500       90
   203  Susan       Mavris     26-NOV-76  AD_VP     7500       40
   202  Pat         Fay        25-OCT-89  AC_MGR    6500       20
   201  Michael     Hartstein  19-JAN-99  IT_PROG   6000       20
   200  Jennifer    Whalen     17-SEP-87  AD_ASST   4400       10
   199  Douglas     Grant      13-JAN-98  SH_CLERK  3000       50
   198  Donald      Connell    21-JUN-99  SH_CLERK  2600       50
   101  Neena       Kochar     10-JUL-03  SH_CLERK  3400       30
   100  Steven      King       15-JUN-02  AD_ASST   8956      100
```

1. First names in ascending order

SQL

```
SELECT first_name AS "Employee_First_Name"
FROM   emp
ORDER  BY first_name ASC;
```

OUTPUT

```
EMPLOYEE_FIRST_NAME
-------------------
Donald
Douglas
Hermann
Jennifer
Michael
Neena
Pat
Shelley
Steven
Susan
William

11 rows selected.
```

2. First names in descending order

SQL

```
SELECT first_name AS "Employee_First_Name"
FROM   emp
ORDER  BY first_name DESC;
```

OUTPUT

```
EMPLOYEE_FIRST_NAME
-------------------
William
Susan
Steven
Shelley
Pat
Neena
Michael
Jennifer
Hermann
Douglas
Donald

11 rows selected.
```

3. Hire dates in ascending order

SQL

```
SELECT hire_date FROM emp ORDER BY hire_date ASC;
```

OUTPUT

```
HIRE_DATE
---------
26-NOV-76
17-SEP-87
25-OCT-89
23-AUG-95
13-JAN-98
24-FEB-98
19-JAN-99
21-JUN-99
12-MAR-01
15-JUN-02
10-JUL-03

11 rows selected.
```

4. Employees whose first name starts with J or M, sorted by last name

SQL

```
SELECT * FROM emp
WHERE  first_name LIKE 'J%' OR first_name LIKE 'M%'
ORDER  BY last_name;
```

OUTPUT

```
EMP_ID  FIRST_NAME  LAST_NAME  HIRE_DATE  JOB_ID    SAL  DEPT_ID
------  ----------  ---------  ---------  -------  ----  -------
   201  Michael     Hartstein  19-JAN-99  IT_PROG  6000       20
   200  Jennifer    Whalen     17-SEP-87  AD_ASST  4400       10

2 rows selected.
```

5. Highest, lowest, average and sum of salary

SQL

```
SELECT MAX(sal) AS "Max", MIN(sal) AS "Min",
       AVG(sal) AS "Avg", SUM(sal) AS "Sum"
FROM   emp;
```

OUTPUT

```
 Max   Min         Avg    Sum
----  ----  ----------  -----
9500  2300  5377.81818  59156

1 row selected.
```

6. Highest, lowest, average and sum of salary for each job type

SQL

```
SELECT job_id, MAX(sal) AS "Max", MIN(sal) AS "Min",
       AVG(sal) AS "Avg", SUM(sal) AS "Sum"
FROM   emp
GROUP  BY job_id;
```

OUTPUT

```
JOB_ID     Max   Min   Avg    Sum
--------  ----  ----  ----  -----
AC_MGR    6500  2300  4400   8800
AD_ASST   8956  4400  6678  13356
AD_PRES   9500  9500  9500   9500
AD_VP     7500  7500  7500   7500
IT_PROG   6000  5000  5500  11000
SH_CLERK  3400  2600  3000   9000

6 rows selected.
```

Note: Oracle returns GROUP BY rows in no guaranteed order; they are shown sorted by JOB_ID here.

7. Number of people under each job

SQL

```
SELECT job_id, COUNT(*) AS no_of_employees
FROM   emp
GROUP  BY job_id;
```

OUTPUT

```
JOB_ID    NO_OF_EMPLOYEES
--------  ---------------
AC_MGR                  2
AD_ASST                 2
AD_PRES                 1
AD_VP                   1
IT_PROG                 2
SH_CLERK                3

6 rows selected.
```

8. Number of managers (without listing ids or names)

SQL

```
SELECT COUNT(*) AS no_of_managers
FROM   emp
WHERE  job_id LIKE '%MGR';
```

OUTPUT

```
NO_OF_MANAGERS
--------------
             2

1 row selected.
```

Note: Managers are identified by job_id ending in MGR (AC_MGR).

9. Difference between highest and lowest salary

SQL

```
SELECT MAX(sal) - MIN(sal) AS difference FROM emp;
```

OUTPUT

```
DIFFERENCE
----------
      7200

1 row selected.
```

10. Maximum and average salary of the engineers

SQL

```
SELECT MAX(sal) AS max_sal, AVG(sal) AS avg_sal
FROM   emp
WHERE  job_id = 'IT_PROG';
```

OUTPUT

```
MAX_SAL  AVG_SAL
-------  -------
   6000     5500

1 row selected.
```

Note: The table has no 'engineer' job; IT_PROG (programmers) is the closest match.

11. First and last first name in the alphabetised list

SQL

```
SELECT MIN(first_name) AS first_name_first,
       MAX(first_name) AS first_name_last
FROM   emp;
```

OUTPUT

```
FIRST_NAME_FIRST  FIRST_NAME_LAST
----------------  ---------------
Donald            William

1 row selected.
```

12. First and last hire dates

SQL

```
SELECT MIN(hire_date) AS "First Hire_Date",
       MAX(hire_date) AS "Last Hire_Date"
FROM   emp;
```

OUTPUT

```
First Hire_Date  Last Hire_Date
---------------  --------------
26-NOV-76        10-JUL-03

1 row selected.
```

13. Maximum and average salary of the clerks

SQL

```
SELECT MAX(sal) AS max_sal, AVG(sal) AS avg_sal
FROM   emp
WHERE  job_id LIKE '%CLERK';
```

OUTPUT

```
MAX_SAL  AVG_SAL
-------  -------
   3400     3000

1 row selected.
```

14. Department number and lowest salary of each department

SQL

```
SELECT dept_id, MIN(sal) AS min_salary
FROM   emp
GROUP  BY dept_id;
```

OUTPUT

```
DEPT_ID  MIN_SALARY
-------  ----------
     10        4400
     20        6000
     30        3400
     40        7500
     50        2600
     60        2300
     90        9500
    100        8956

8 rows selected.
```

15. Same, excluding groups where the minimum salary is 3000 or less, highest first

SQL

```
SELECT dept_id, MIN(sal) AS min_salary
FROM   emp
GROUP  BY dept_id
HAVING MIN(sal) > 3000
ORDER  BY min_salary DESC;
```

OUTPUT

```
DEPT_ID  MIN_SALARY
-------  ----------
     90        9500
    100        8956
     40        7500
     20        6000
     10        4400
     30        3400

6 rows selected.
```

16. Sentence format for every row

SQL

```
SELECT first_name || ' whose designation is ' || job_id ||
       ' gets ' || sal || ' but wants to earn ' || 3*sal || '.' AS statement
FROM   emp;
```

OUTPUT

```
STATEMENT
--------------------------------------------------------------------
Donald whose designation is SH_CLERK gets 2600 but wants to earn 7800.
Douglas whose designation is SH_CLERK gets 3000 but wants to earn 9000.
Jennifer whose designation is AD_ASST gets 4400 but wants to earn 13200.
Michael whose designation is IT_PROG gets 6000 but wants to earn 18000.
Pat whose designation is AC_MGR gets 6500 but wants to earn 19500.
Susan whose designation is AD_VP gets 7500 but wants to earn 22500.
Hermann whose designation is AD_PRES gets 9500 but wants to earn 28500.
Shelley whose designation is AC_MGR gets 2300 but wants to earn 6900.
William whose designation is IT_PROG gets 5000 but wants to earn 15000.
Steven whose designation is AD_ASST gets 8956 but wants to earn 26868.
Neena whose designation is SH_CLERK gets 3400 but wants to earn 10200.

11 rows selected.
```

17. Employee details as a comma-separated line

SQL

```
SELECT emp_id || ', ' || first_name || ', ' || last_name || ', ' ||
       TO_CHAR(hire_date,'DD-Mon-YY') || ', ' || job_id || ', ' ||
       sal || ', ' || dept_id AS emp_details
FROM   emp;
```

OUTPUT

```
EMP_DETAILS
-----------------------------------------------------------
198, Donald, Connell, 21-Jun-99, SH_CLERK, 2600, 50
199, Douglas, Grant, 13-Jan-98, SH_CLERK, 3000, 50
200, Jennifer, Whalen, 17-Sep-87, AD_ASST, 4400, 10
201, Michael, Hartstein, 19-Jan-99, IT_PROG, 6000, 20
202, Pat, Fay, 25-Oct-89, AC_MGR, 6500, 20
203, Susan, Mavris, 26-Nov-76, AD_VP, 7500, 40
204, Hermann, Baer, 23-Aug-95, AD_PRES, 9500, 90
205, Shelley, Higgins, 24-Feb-98, AC_MGR, 2300, 60
206, William, Gitz, 12-Mar-01, IT_PROG, 5000, 60
100, Steven, King, 15-Jun-02, AD_ASST, 8956, 100
101, Neena, Kochar, 10-Jul-03, SH_CLERK, 3400, 30

11 rows selected.
```

18. Today's date as TODAY

SQL

```
SELECT SYSDATE AS today FROM dual;
```

OUTPUT

```
TODAY
---------
09-NOV-26

1 row selected.
```

Note: SYSDATE is the date the script was run: 09-Nov-2026.

19. Employee id, day and year of hiring

SQL

```
SELECT emp_id, TO_CHAR(hire_date,'Day') AS day_hired,
       TO_CHAR(hire_date,'YYYY') AS year_hired
FROM   emp;
```

OUTPUT

```
EMP_ID  DAY_HIRED  YEAR_HIRED
------  ---------  ----------
   198  Monday     1999
   199  Tuesday    1998
   200  Thursday   1987
   201  Tuesday    1999
   202  Wednesday  1989
   203  Friday     1976
   204  Wednesday  1995
   205  Tuesday    1998
   206  Monday     2001
   100  Saturday   2002
   101  Thursday   2003

11 rows selected.
```

20. Employee name and hire date in 'dd-month-yy'

SQL

```
SELECT first_name || ' ' || last_name AS name,
       TO_CHAR(hire_date,'dd-month-yy') AS hire_date
FROM   emp;
```

OUTPUT

```
NAME               HIRE_DATE
-----------------  ---------------
Donald Connell     21-june     -99
Douglas Grant      13-january  -98
Jennifer Whalen    17-september-87
Michael Hartstein  19-january  -99
Pat Fay            25-october  -89
Susan Mavris       26-november -76
Hermann Baer       23-august   -95
Shelley Higgins    24-february -98
William Gitz       12-march    -01
Steven King        15-june     -02
Neena Kochar       10-july     -03

11 rows selected.
```

Note: Oracle pads the month name to 9 characters; use 'dd-fmmonth-yy' to remove the padding.

21. Employee id and month of hiring

SQL

```
SELECT emp_id, TO_CHAR(hire_date,'Month') AS month_hired FROM emp;
```

OUTPUT

```
EMP_ID  MONTH_HIRED
------  -----------
   198  June
   199  January
   200  September
   201  January
   202  October
   203  November
   204  August
   205  February
   206  March
   100  June
   101  July

11 rows selected.
```

22. Name, id and hire date in 'dd-mon-yyyy'

SQL

```
SELECT first_name || ' ' || last_name AS name, emp_id,
       TO_CHAR(hire_date,'dd-mon-yyyy') AS hire_date
FROM   emp;
```

OUTPUT

```
NAME               EMP_ID  HIRE_DATE
-----------------  ------  -----------
Donald Connell        198  21-jun-1999
Douglas Grant         199  13-jan-1998
Jennifer Whalen       200  17-sep-1987
Michael Hartstein     201  19-jan-1999
Pat Fay               202  25-oct-1989
Susan Mavris          203  26-nov-1976
Hermann Baer          204  23-aug-1995
Shelley Higgins       205  24-feb-1998
William Gitz          206  12-mar-2001
Steven King           100  15-jun-2002
Neena Kochar          101  10-jul-2003

11 rows selected.
```

23. Name, id and hire date in 'month-dd-yyyy'

SQL

```
SELECT first_name || ' ' || last_name AS name, emp_id,
       TO_CHAR(hire_date,'month-dd-yyyy') AS hire_date
FROM   emp;
```

OUTPUT

```
NAME               EMP_ID  HIRE_DATE
-----------------  ------  -----------------
Donald Connell        198  june     -21-1999
Douglas Grant         199  january  -13-1998
Jennifer Whalen       200  september-17-1987
Michael Hartstein     201  january  -19-1999
Pat Fay               202  october  -25-1989
Susan Mavris          203  november -26-1976
Hermann Baer          204  august   -23-1995
Shelley Higgins       205  february -24-1998
William Gitz          206  march    -12-2001
Steven King           100  june     -15-2002
Neena Kochar          101  july     -10-2003

11 rows selected.
```

Note: Month is blank-padded to 9 characters by Oracle; 'fmmonth-dd-yyyy' removes the padding.

24. System year in full spelling

SQL

```
SELECT TO_CHAR(TO_DATE(TO_CHAR(SYSDATE,'YYYY'),'J'),'Jsp') AS year_in_words
FROM   dual;
```

OUTPUT

```
YEAR_IN_WORDS
-----------------------
Two Thousand Twenty-Six

1 row selected.
```

Note: Format 'J' is the Julian day number, so TO_DATE(year,'J') followed by 'Jsp' spells the number (1999 would give One Thousand Nine Hundred Ninety-Nine).

25. Dates 15 days after and 15 days before today

SQL

```
SELECT SYSDATE + 15 AS after_15_days,
       SYSDATE - 15 AS before_15_days
FROM   dual;
```

OUTPUT

```
AFTER_15_DAYS  BEFORE_15_DAYS
-------------  --------------
24-NOV-26      25-OCT-26

1 row selected.
```

26. Ensure the domain and entity integrity constraints

SQL

```
-- constraints defined on EMP
SELECT constraint_name, constraint_type, search_condition
FROM   user_constraints WHERE table_name = 'EMP';

-- entity integrity: duplicate primary key
INSERT INTO EMP VALUES (100,'Test','User',SYSDATE,'IT_PROG',1000,20);
-- entity integrity: NULL primary key
INSERT INTO EMP VALUES (NULL,'Test','User',SYSDATE,'IT_PROG',1000,20);
-- domain integrity: DEPT_ID below 10
INSERT INTO EMP VALUES (300,'Test','User',SYSDATE,'IT_PROG',1000,5);
```

OUTPUT

```
SQL> SELECT constraint_name, constraint_type, search_condition
     FROM user_constraints WHERE table_name = 'EMP';

CONSTRAINT_NAME  C  SEARCH_CONDITION
---------------  -  ----------------------
PK_EMP           P  EMP_ID
NN_EMP_FNAME     C  FIRST_NAME IS NOT NULL
NN_EMP_LNAME     C  LAST_NAME IS NOT NULL
NN_EMP_HDATE     C  HIRE_DATE IS NOT NULL
NN_EMP_JOB       C  JOB_ID IS NOT NULL
NN_EMP_SAL       C  SAL IS NOT NULL
CHK_EMP_DEPT     C  DEPT_ID >= 10

7 rows selected.

-- duplicate EMP_ID 100:
ERROR at line 1:
ORA-00001: unique constraint (PK_EMP) violated

-- NULL EMP_ID:
ERROR at line 1:
ORA-01400: cannot insert NULL into ("EMP"."EMP_ID")

-- DEPT_ID = 5:
ERROR at line 1:
ORA-02290: check constraint (CHK_EMP_DEPT) violated
```

Note: C = check/not-null, P = primary key. The primary key stops duplicate and NULL EMP_ID values (entity integrity); the CHECK on DEPT_ID restricts it to values >= 10 (domain integrity). All rejected rows were rolled back.

27. Drop the domain constraint

SQL

```
ALTER TABLE EMP DROP CONSTRAINT chk_emp_dept;
```

OUTPUT

```
Table altered.

SQL> SELECT constraint_name, constraint_type, search_condition
     FROM user_constraints WHERE table_name = 'EMP';

CONSTRAINT_NAME  C  SEARCH_CONDITION
---------------  -  ----------------------
PK_EMP           P  EMP_ID
NN_EMP_FNAME     C  FIRST_NAME IS NOT NULL
NN_EMP_LNAME     C  LAST_NAME IS NOT NULL
NN_EMP_HDATE     C  HIRE_DATE IS NOT NULL
NN_EMP_JOB       C  JOB_ID IS NOT NULL
NN_EMP_SAL       C  SAL IS NOT NULL

6 rows selected.
```

28. Drop the primary key

SQL

```
ALTER TABLE EMP DROP PRIMARY KEY;
-- or: ALTER TABLE EMP DROP CONSTRAINT pk_emp;
```

OUTPUT

```
Table altered.

SQL> SELECT constraint_name, constraint_type, search_condition
     FROM user_constraints WHERE table_name = 'EMP';

CONSTRAINT_NAME  C  SEARCH_CONDITION
---------------  -  ----------------------
NN_EMP_FNAME     C  FIRST_NAME IS NOT NULL
NN_EMP_LNAME     C  LAST_NAME IS NOT NULL
NN_EMP_HDATE     C  HIRE_DATE IS NOT NULL
NN_EMP_JOB       C  JOB_ID IS NOT NULL
NN_EMP_SAL       C  SAL IS NOT NULL

5 rows selected.
```

29. Drop the table

SQL

```
DROP TABLE EMP;
SELECT * FROM EMP;
```

OUTPUT

```
Table dropped.

SELECT * FROM EMP
              *
ERROR at line 1:
ORA-00942: table or view does not exist
```